TBO Inventory Intelligence — Project Report

Nov 12, 2026 · @srujan · Team Void

Submission links

Material	Link
Live prototype (Vercel)	tbo-inventory-optimizer-sggo.vercel.app
Source code (GitHub)	github.com/srujanmh/tbo-inventory-optimizer
Video pitch, no voice, on-screen captions (3.5 min)	docs/TBO_Inventory_Intelligence_pitch_silent.mp4
10-slide presentation	Pitch deck

All results in this report come from a synthetic dataset. They show how the method performs; they are not actual TBO revenue, profit or savings.

Executive summary

TBO Inventory Intelligence takes a maximum budget and returns a pre-buy plan: which hotel rooms and flight seats to buy, how many, from which supplier, by when, and at what selling price. It maximises expected profit while protecting the worst 10% of outcomes.

Result (synthetic data, ₹75 Cr budget)	Value
Weekly demand forecast accuracy (1 − WAPE, 13 unseen weeks)	87.5% (naive baseline 81.3%)
Expected profit of the plan (forecast)	₹13.53 Cr, ROI 18.1%
Expected uplift vs repeating last year's mix (forecast)	+28.5% (+₹3.00 Cr)
Saved by buying at the recommended lead time	₹7.6 Cr
Back-test: realised profit of a plan made on 11 Aug 2026	₹11.74 Cr, within 2.6% of its prediction
Back-test: realised uplift vs last year's mix	+61%
Independent maths checks passed	24 of 24

The expected uplift is a forecast for the coming quarter. The back-test is a separate measurement of what a past plan actually earned. They are reported separately throughout.

Problem and objectives

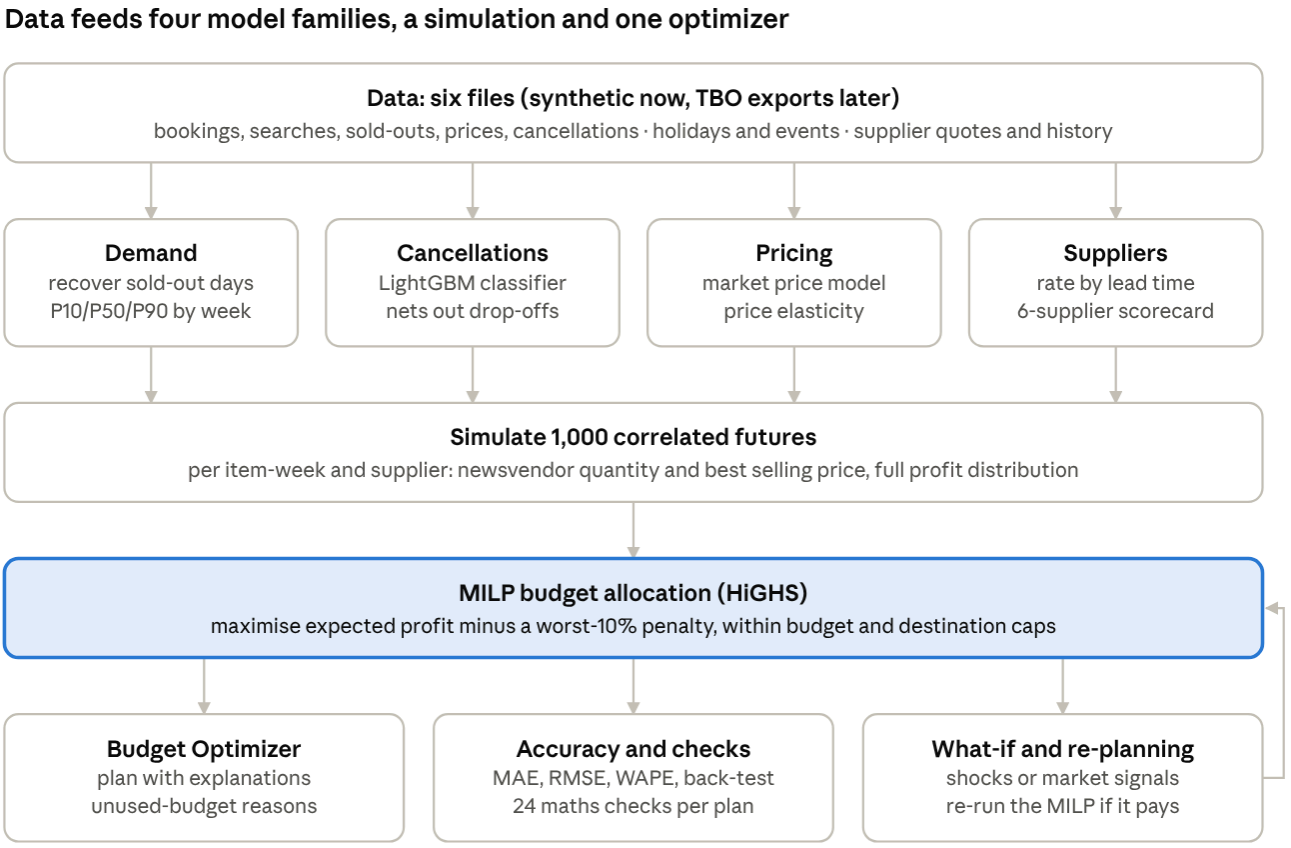
TBO can pre-buy rooms and seats below spot rates and secure supply before it sells out, but every pre-buy is a bet: too little loses bookings, too much leaves unsold stock that is only partly refundable.

Three things make it hard on our dataset:

- On 27% of days inventory is sold out, so recorded bookings understate real demand.
- 585 hotel-weeks and route-weeks compete for one budget, each with different margin, demand and risk.
- Only 15–60% of unsold stock is refundable, and bookings cancel.

Objective in the brief	How the prototype delivers it
Predict demand	Quantile forecasts from bookings, searches, holidays and events; hidden demand recovered on sold-out days
Identify opportunities	532 profitable item-weeks scored on ROI, sold-out pressure and forecast confidence
Optimize investments	MILP allocates the budget across hotels, flights, destinations, weeks and suppliers
Maximize profitability	Quantity, buy-by date, supplier and selling price chosen per position
Manage risk	1,000 simulated futures, a worst-10% (CVaR) term in the objective, destination caps, a risk slider

Solution architecture



solution architecture · data to plan

Six data files train four model families; their outputs drive a 1,000-scenario simulation, and one MILP turns those distributions into the plan. The what-if page and the automated re-planner feed changed conditions back into the same optimizer.

Models and techniques

Component	Technique	Role in the plan
Demand unconstraining	Searches ÷ look-to-book ratio on sold-out days	Recovers demand that sold-out days hide
Demand forecast	LightGBM quantile regression (P10/P50/P90), log target, seed bagging, blended with a direct weekly model, calibrated intervals	Demand distribution per item and travel week

Component	Technique	Role in the plan
Cancellations	LightGBM classifier	Expected cancellations reduce sellable demand in every scenario
Price response	Log-log elasticity regression on unconstrained demand	How demand moves with the selling price
Market price	LightGBM, holiday- and event-aware	Selling price benchmark per week
Supplier rates	LightGBM on lead time	Cheapest moment to buy (buy-by date)
Supplier scorecard	Estimated from allotment history	Price vs benchmark, availability, confirmation rate, refund rate, minimum notice per supplier
Simulation	1,000 correlated Monte Carlo scenarios (market, destination and item shocks)	Profit distribution for every option
Quantity	Newsvendor critical ratio over a price grid	Units and price per item-week and supplier
Allocation	Mixed-integer program (PuLP + HiGHS), mean minus tail-risk objective, destination caps	Splits the budget across all positions

No model uses the generator's hidden parameters as inputs, and every lagged feature is older than the 13-week forecast horizon, so there is no leakage.

Accuracy and validation

Weekly demand forecasts are 87.5% accurate on 13 weeks the model never saw (11 Aug → 9 Nov 2026), against 81.3% for the same-week-last-year baseline. That is the only figure called "accuracy" anywhere in the product.

Accuracy = 1 − WAPE, where WAPE = Σ|forecast − actual| ÷ Σ actual over every item and week in the holdout. Forecasts for each week use only data up to the day before the holdout starts.

Forecast (holdout)	Accuracy	WAPE	MAE (units)	RMSE (units)	Bias	Points
Weekly, our model	**87.5%**	12.5%	34.3	56.6	−1.2%	540

Forecast (holdout)	Accuracy	WAPE	MAE (units)	RMSE (units)	Bias	Points
Weekly, naive (same week last year)	81.3%	18.7%	51.1	88.0	−9.3%	540
Weekly, best achievable (noise floor)	89.8%	10.2%	28.0	44.9	−0.5%	540
Weekly, hotels only	85.9%	14.1%	22.5	32.6	−3.4%	360
Weekly, flights only	88.5%	11.5%	57.9	86.5	+0.2%	180
Daily, our model	71.4%	28.6%	11.2	18.8	−1.8%	4,095
Daily, naive	59.8%	40.2%	15.8	26.7	−9.9%	4,095

Forecast ranges are calibrated: 80% of actual weekly values fall inside the P10–P90 range (target 80%). Supporting models are scored the same way: recovered demand on sold-out days is within 9.0% of the truth (raw bookings: 36.3%), supplier-rate predictions have 1.8% MAPE (baseline 6.5%), and weekly market prices have 2.6% MAPE.

The gap to the 89.8% noise floor is 2.3 points, and forecast errors do not grow with how far ahead the week is, so most of what remains is randomness no model can predict. The Accuracy & Validation page plots predicted against actual demand for every item.

Profit: expected results and back-test

Two different measurements, reported separately.

Expected results (forecast for 16 Nov 2026 → 14 Feb 2027). The plan for a ₹75 Cr budget, averaged over 1,000 simulated futures:

Strategy, same ₹75 Cr budget	Expected profit	Worst-10% average profit	ROI
AI optimizer (MILP)	**₹13.53 Cr**	**₹12.30 Cr**	**18.1%**
Greedy ROI ranking	₹13.32 Cr	₹11.01 Cr	17.8%
Last year's mix	₹10.53 Cr	₹9.36 Cr	15.6%

Back-test (what a past plan actually earned). Every model was re-fitted using only data available on 11 Aug 2026, a plan was made for the next 12 travel weeks, and it was replayed against what then happened: real demand, cancellations, supplier quotes, and each supplier's real confirmation failures and refunds.

Strategy, same ₹75 Cr budget	Invested	Predicted profit	Realised profit	Realised ROI	Loss-making positions
AI optimizer (MILP)	₹64.1 Cr	₹12.04 Cr	**₹11.74 Cr**	**18.3%**	3 of 495
Greedy ROI ranking	₹66.7 Cr	₹12.10 Cr	₹11.98 Cr	18.0%	4

Strategy, same ₹75 Cr budget	Invested	Predicted profit	Realised profit	Realised ROI	Loss-making positions
Last year's mix	₹68.8 Cr	₹7.63 Cr	₹7.30 Cr	10.6%	95

Realised profit landed within 2.6% of the prediction and 61% above last year's mix. The greedy ranking earned ₹0.24 Cr more but tied up ₹2.6 Cr more capital, so the optimizer returns more per rupee. The optimizer left ₹10.9 Cr unspent because no further unit added risk-adjusted profit.

Product features

The prototype is a web app with nine pages. A FastAPI backend runs the full MILP; a static build runs a matching in-browser engine.

Page	What the user can do
Budget Optimizer	Set budget, risk appetite, destination cap, inventory types and destinations; get the plan, KPIs, three-strategy comparison, profit distribution and allocation by destination and week. When budget is left unused, a card explains why.
Plan explanations	Click any position: quantity logic (newsvendor ratio), buy-by date, selling price, cancellation risk, demand drivers, expected profit and risk, and a side-by-side comparison of the eligible suppliers.
What-if & Re-planning	Change demand, market prices, supplier costs, supplier availability or cancellations for any destination, item type or supplier; see the new plan, the positions that change, and the value of re-planning versus keeping the current plan.
Automated re-planning	A feed of market signals (simulated in the demo); after each, the system re-optimises and adopts the new plan only if it beats the current one by at least 0.5% of expected profit, with a log of every decision.
Suppliers	Scorecard of 6 suppliers: price vs benchmark, availability, confirmation rate, refund on release, minimum notice, free-release window, and how many item-weeks each wins.

Page	What the user can do
Demand & Opportunities	Destination × week heatmap and ranked opportunities.
Forecast Explorer	History, recovered demand and the 13-week forecast with ranges for every item.
Accuracy & Validation	Metric definitions, the error table, predicted vs actual charts, the back-test and the live maths checks.
Risk & Budget Strategy	Risk–return frontier and the profit curve across budgets.
Data Sources	The data contract, empty templates and a file checker for real data.

Mathematical verification

All 24 independent checks pass on the ₹75 Cr plan. Each recomputes a number from first principles instead of calling the code that produced it, and the app re-runs them on whatever plan is on screen.

Group	What is checked	Result on the ₹75 Cr plan
Budget	Budget respected, positions add up, destination cap, unused = budget − invested	₹74.98 Cr invested of ₹75 Cr; largest destination ₹14.6 Cr under the ₹22.5 Cr cap
Profit	505 position profits recomputed from scratch; portfolio total; ROI	max relative difference 5.5×10^{-8}
Risk	Worst-10% profit, 5th percentile, probability of loss, profit spread	all match to the rupee
Inventory	Units add up, expected sales ⩽ units bought, sell-through, quantity blocks, newsvendor quantities	exact
Optimality	Beats the greedy plan on its own objective; distance from the LP upper bound	within 0.009% of the theoretical best
Forecast	WAPE, MAE, RMSE and accuracy recomputed; model beats naive	12.5% vs naive 18.7%

The same logic runs as 8 automated tests (`cd backend && pytest tests`) and at `POST /api/verify` .

Data, limitations and next steps

The challenge supplied no data, so the prototype runs on a synthetic dataset: 45 items (30 hotels in 10 destinations and 3 tiers, plus 15 flight routes), daily from 1 Jan 2024 to 9 Nov 2026, with seasonality, Indian holidays and destination events, sold-out days, cancellations, lead-time supplier quotes and 6 suppliers with different terms. Hidden true values are used only to score the models.

Moving to real data needs no model changes. Every model reads only six files:

1. Export `items` , `daily` , `calendar` , `rate_quotes` , `suppliers` and `supplier_history` (at least 15 months) from TBO's booking, search, contracting and procurement systems. The Data Sources page lists every column and rule and offers empty templates.
2. Validate and load: `python -m pipeline.ingest --source /path/to/export` . Nothing is copied unless every table passes.
3. Retrain: `python -m pipeline.train` (about 2 minutes). It rebuilds every model and re-runs both back-tests and the maths checks.

Limitations

- All profit, savings and back-test figures are simulated; they show the method, not TBO's results.
- The re-planning feed uses simulated market signals; live search, competitor-rate and supplier-availability feeds still need to be connected.
- Forecast accuracy is close to the noise floor of this dataset; on real data, searches for future travel dates are the main lever for further gains.
- The cancellation model only matches the per-item baseline, because cancellations in this dataset barely vary beyond the item.

Next steps

- Load TBO's data through the contract and retrain.
- Connect live feeds to the automated re-planner.
- Add forward-looking searches as a forecast feature.
- Connect supplier APIs for one-click allotment purchase, and model pre-buy as a top-up to dynamic sourcing.